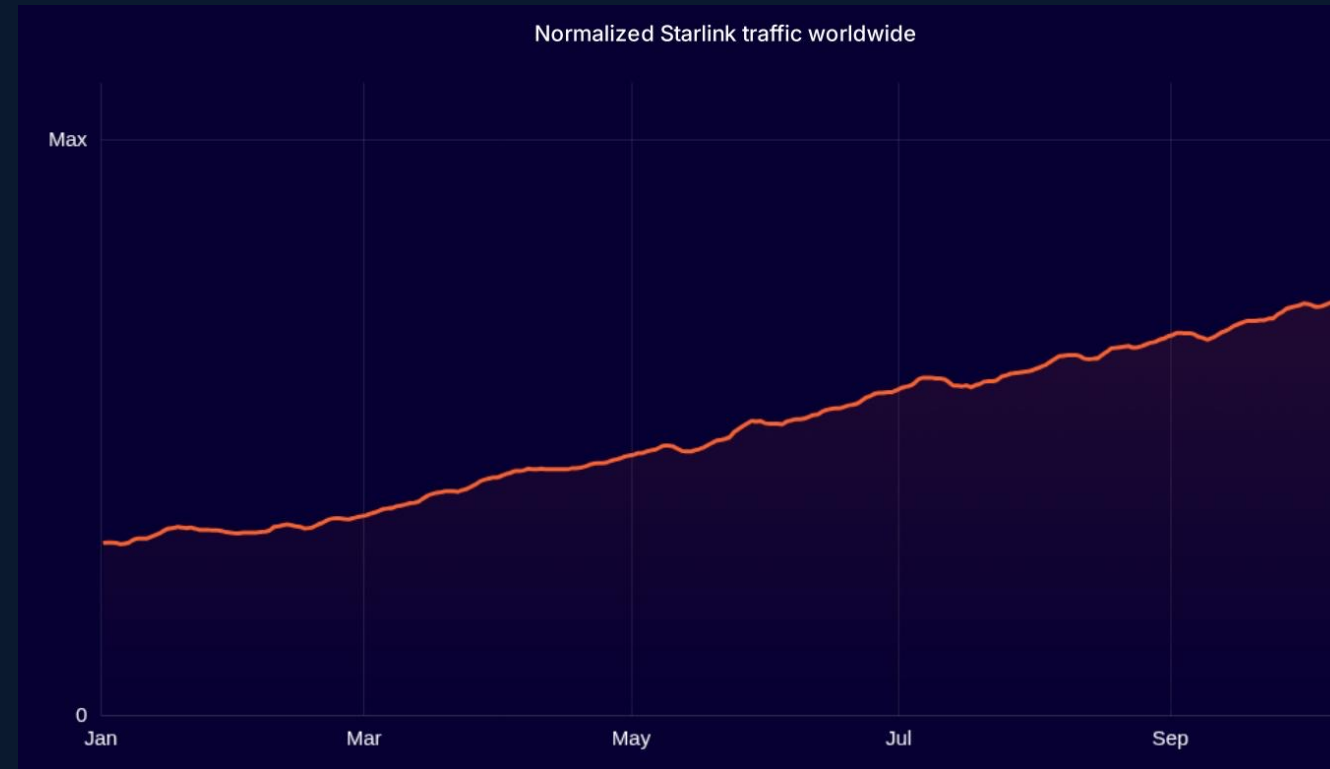


Cloudflare Radar 2024 Year in Review

a-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security th

17% Global Traffic Growth and Starlink's 3.3x Surge

Global Internet traffic grew by 17% in 2024, continuing the upward trend driven by increasing reliance on the Internet for communication, commerce, entertainment, and transportation. SpaceX Starlink traffic saw dramatic 3.3x growth during 2024. Starlink expands satellite connectivity to previously underserved locations. Trend line scaled relative to peak observed request volume.



Google Leads Top 10 Services as Generative AI Rises

Top 10 Most Popular Internet Services

(2024):

1. Google

2. Facebook

3. Apple

4. TikTok

5. AWS
6. Microsoft

7. Instagram

8. YouTube

9. Amazon

10. WhatsApp

Key Trends:

Google maintains its #1 position for the second consecutive year.

Generative AI continues its rapid and meteoric rise.

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Post-Quantum Encryption Jumps from 2% to 13% of TLS

1.3

Post-Quantum Encryption Surges:

Jumped from 2% to ~13% of TLS 1.3 traffic in 2024
Driven by Google Chrome 124 enabling PQ key agreement by default in April

Other Technology Shifts:

HTTP/3 Adoption: Reached 21% of total traffic

API Client Languages:

- Go is the top language (~12% of automated requests)

- Followed by Node.js, Python (9.6%), Java (7.4%), .NET (3.6%)

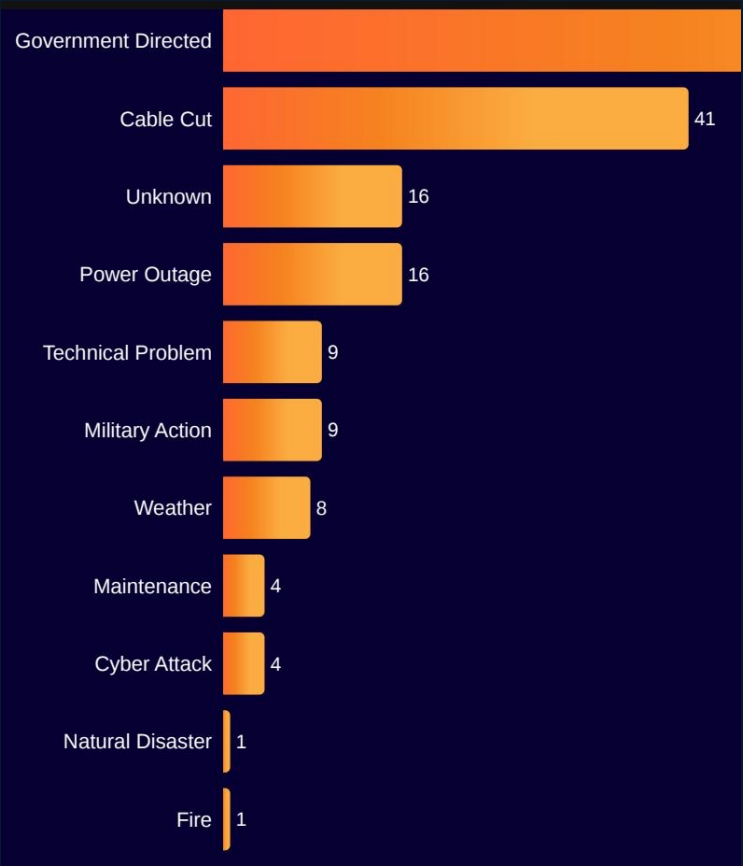


225 Major Disruptions: Government Shutdowns Lead Outage Causes

Cloudflare observed 225 major Internet disruptions globally in 2024. Government-directed shutdowns remain the leading cause of outages.

Disruption Causes:

- Government Directed: 116+
- Cable Cut: 41
- Unknown: 16
- Power Outage: 16
- Technical Problem: 9
- Military Action: 9
- Weather: 8
- Maintenance: 4
- Cyber Attack: 4
- Natural Disaster: 1
- Fire: 1



6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Overall Mitigation & Attack Trends:

6.5% of all processed traffic was mitigated, indicating massive scale of ongoing threats.

Gambling and Games was the most attacked industry category.

68% of global bot traffic originated from just the top 10 countries.

Persistent Vulnerabilities & Email Threats:

Log4j (CVE-2021-44228) continues to be actively targeted more than two years later.

4.3% of processed emails were malicious; 43% of those contained deceptive links.

The .bar TLD originated the largest share of malicious and spam emails.

Five Takeaways Shaping the 2025 Internet Outlook

1. Unstoppable Traffic Growth Trajectory

17% global growth paired with rapid satellite (Starlink 3.3x) expansion into new regions.

2. Accelerated Encryption Momentum

Post-quantum encryption leaped to 13%, driven by browser defaults, securing the future.

3. Evolving Developer Ecosystem

Go has emerged as the top API client language, alongside 21% HTTP/3 adoption.

4. Persistent Security Threats

6.5% traffic mitigated; aging flaws like Log4j remain prime targets for attackers.

5. Connectivity Fragility from Government Actions

With 116+ incidents, government-directed shutdowns are the leading cause of major disruptions.